DAY 26 – DATA PRIVACY BASICS

Cyber Security Internship – Veda Technology

1. Objective

To understand the basic principles of data privacy, identify different types of personal and organizational information, and classify data according to its sensitivity. The task focuses on categorizing information as public, internal, confidential, or sensitive and recommending appropriate protection measures to prevent unauthorized access and data misuse.

2. Introduction

Data privacy refers to the responsible collection, processing, storage, sharing, and protection of personal information. It ensures that individuals have appropriate control over how their information is used and that organizations handle data according to applicable privacy requirements.

Organizations collect different types of information, including customer names, contact details, financial records, employee information, and authentication credentials.

Not all information requires the same level of protection. Data classification helps organizations identify the sensitivity of information and apply appropriate security controls.

Proper data privacy practices reduce the risk of data breaches, identity theft, unauthorized disclosure, and misuse of personal information.

3. Fundamental Principles of Data Privacy

Principle	Description
Data Minimization	Collect only the information necessary for a specific purpose
Purpose Limitation	Use personal data only for legitimate, specified purposes
Access Control	Restrict access to authorized individuals
Transparency	Inform individuals about how their data is processed
Data Accuracy	Maintain accurate and updated information
Storage Limitation	Retain personal data only as long as necessary
Confidentiality	Protect information from unauthorized disclosure
Accountability	Maintain responsibility for appropriate data handling

4. Data Classification Categories

Data classification involves organizing information based on its sensitivity, intended audience, and potential impact if exposed.

Classification	Description	Examples
Public	Information approved for general access	Company website, published announcements
Internal	Information intended for organizational use	Internal schedules, department reports
Confidential	Information requiring restricted access	Customer contact details, business contracts
Sensitive	Highly protected information with significant privacy or security implications	Passwords, financial credentials, government identifiers

5. Tools and Dataset Description

Google Sheets is used to organize and classify synthetic customer information.

The sample dataset contains fictional customer records and organizational information to demonstrate data classification.

Tool	Purpose
Google Sheets	Creating and organizing data classification tables
Data Filters	Sorting information by classification
Conditional Formatting	Highlighting different sensitivity levels
Synthetic Dataset	Practicing classification without exposing real customer data

6. Practical Implementation

Step 1: Prepare Synthetic Customer Data

A sample dataset is created using fictional customer details and organizational records.

Record ID	Information Type	Sample Value
D01	Customer Name	Ananya Rao
D02	Email Address	ananya@example.com
D03	Contact Number	9000000000
D04	Customer ID	CUST1001
D05	Internal Employee Schedule	Shift A – 9 AM
D06	Company Website	www.example.com

D07	Bank Account Number	XXXX XXXX 1234
D08	Account Password	[REDACTED]
D09	Company Announcement	Product Launch
D10	Customer Purchase History	Order #1024
D11	Government ID Number	[REDACTED]
D12	Internal Sales Report	Monthly Revenue Summary

Step 2: Classify the Data

Each information type is reviewed and assigned an appropriate classification based on its privacy and security requirements.

Record ID	Information Type	Classification	Reason
D01	Customer Name	Confidential	Identifies an individual
D02	Email Address	Confidential	Personal contact information
D03	Contact Number	Confidential	Can be used to contact or identify a person
D04	Customer ID	Confidential	Links records to a specific customer
D05	Employee Schedule	Internal	Intended for authorized staff
D06	Company Website	Public	Approved for public access
D07	Bank Account Number	Sensitive	Financial information requiring stronger protection
D08	Account Password	Sensitive	Authentication information
D09	Company Announcement	Public	Intended for public communication
D10	Purchase History	Confidential	Reveals customer transaction activity
D11	Government ID Number	Sensitive	Highly identifying personal information
D12	Internal Sales Report	Internal	Intended for organizational use

Step 3: Apply Classification Labels

The dataset is organized in Google Sheets with a separate column for data classification.

Suggested color coding:

Classification	Color	Protection Priority
Public	Green	Low
Internal	Blue	Moderate
Confidential	Yellow	High
Sensitive	Red	Very High

Conditional formatting can be applied to visually distinguish information according to its classification.

Step 4: Identify Privacy Risks

The classified records are reviewed to identify possible risks associated with unauthorized disclosure or improper handling.

Data Type	Potential Privacy Risk
Customer Name	Unwanted identification or profiling
Email Address	Phishing and unwanted communication
Contact Number	Spam, scams, or impersonation
Customer ID	Unauthorized customer record association
Employee Schedule	Exposure of internal operational information
Bank Account Details	Financial fraud
Passwords	Unauthorized account access
Purchase History	Exposure of personal purchasing behavior
Government ID	Identity theft or impersonation
Internal Sales Report	Disclosure of business information

Step 5: Recommend Protection Measures

Suitable security controls are recommended based on the sensitivity of each information category.

Classification	Recommended Protection Measures
Public	Verify accuracy and authorize publication
Internal	Employee-only access and secure organizational storage
Confidential	Role-based access control, encryption, and access monitoring

Sensitive	Strong encryption, MFA, masking, strict access controls, and secure credential handling

Additional Considerations:

- Passwords should be stored using appropriate salted password hashing rather than reversible encryption.
- Financial information and government identifiers should be masked when full values are unnecessary.
- Personal information should be deleted securely when it is no longer required, subject to applicable retention obligations.

7. Data Classification Summary

The synthetic dataset contains 12 records distributed across four classification categories.

Classification	Number of Records	Percentage
Public	2	16.67%
Internal	2	16.67%
Confidential	5	41.67%
Sensitive	3	25.00%
Total	12	100%

Observations

- Confidential information represents the largest portion of the sample dataset.
- Sensitive information requires the strongest security controls.
- Public information can be shared openly after authorization.
- Internal information should remain accessible only to appropriate organizational personnel.
- Classification helps determine suitable access restrictions and protection measures.

8. Data Privacy Risks and Preventive Measures

Privacy Risk	Description	Preventive Measure
Unauthorized Access	Accessing information without permission	Role-based access control
Data Leakage	Accidental or deliberate disclosure	Encryption and secure sharing

Identity Theft	Misuse of personal identifiers	Data masking and minimization
Credential Exposure	Disclosure of passwords or authentication secrets	Secure password hashing and MFA
Excessive Data Collection	Collecting unnecessary personal information	Data minimization
Improper Data Sharing	Sharing data with unauthorized parties	Access reviews and approval procedures
Data Retention Risks	Keeping information longer than necessary	Retention and deletion policies

9. Importance of Data Privacy

Data privacy is an important part of cybersecurity and responsible information management.

1. Protects Personal Information: Reduces the risk of unauthorized disclosure and misuse.
2. Prevents Identity Theft: Helps protect identifying and financial information.
3. Builds Customer Trust: Encourages responsible handling of customer data.
4. Supports Regulatory Compliance: Helps organizations meet applicable privacy obligations.
5. Improves Access Control: Ensures information is accessible only to authorized individuals.
6. Reduces Security Risks: Minimizes exposure to data breaches and unauthorized access.

10. Best Practices for Data Protection

- Collect only necessary personal information.
- Classify data based on sensitivity and business requirements.
- Apply the principle of least privilege.
- Use encryption for confidential and sensitive information.
- Implement multifactor authentication for important accounts.
- Avoid storing plaintext passwords.
- Mask sensitive identifiers in reports and shared documents.
- Regularly review access permissions.
- Establish appropriate data retention and deletion policies.

11. Results and Observations

The synthetic data classification exercise demonstrates how customer and organizational information can be categorized based on privacy and security requirements.

Activity	Result
Dataset Preparation	12 synthetic information records
Classification Categories	Public, Internal, Confidential, Sensitive
Public Records	2
Internal Records	2
Confidential Records	5
Sensitive Records	3
Privacy Risk Assessment	Potential risks identified for each information type
Protection Recommendations	Appropriate controls suggested
Main Deliverable	Data classification table

Key Findings

- Different types of information require different levels of protection.
- Confidential and sensitive records should have stricter access controls.
- Data minimization helps reduce unnecessary privacy risks.
- Classification supports consistent handling and secure information sharing.
- Proper data protection requires both technical controls and organizational policies.

12. Conclusion

The Data Privacy Basics task provided an understanding of how organizations classify and protect personal and business information.

Using a synthetic customer dataset, 12 records were categorized as public, internal, confidential, or sensitive. The classification process helped identify potential privacy risks and recommend appropriate protection measures.

Google Sheets provides a simple method for organizing data, assigning classification labels, and reviewing privacy requirements.

Overall, this task strengthened foundational knowledge of data privacy, information classification, access control, and responsible data handling practices, which are essential for maintaining cybersecurity and protecting sensitive information.